

Reference Material Explorer (RME)

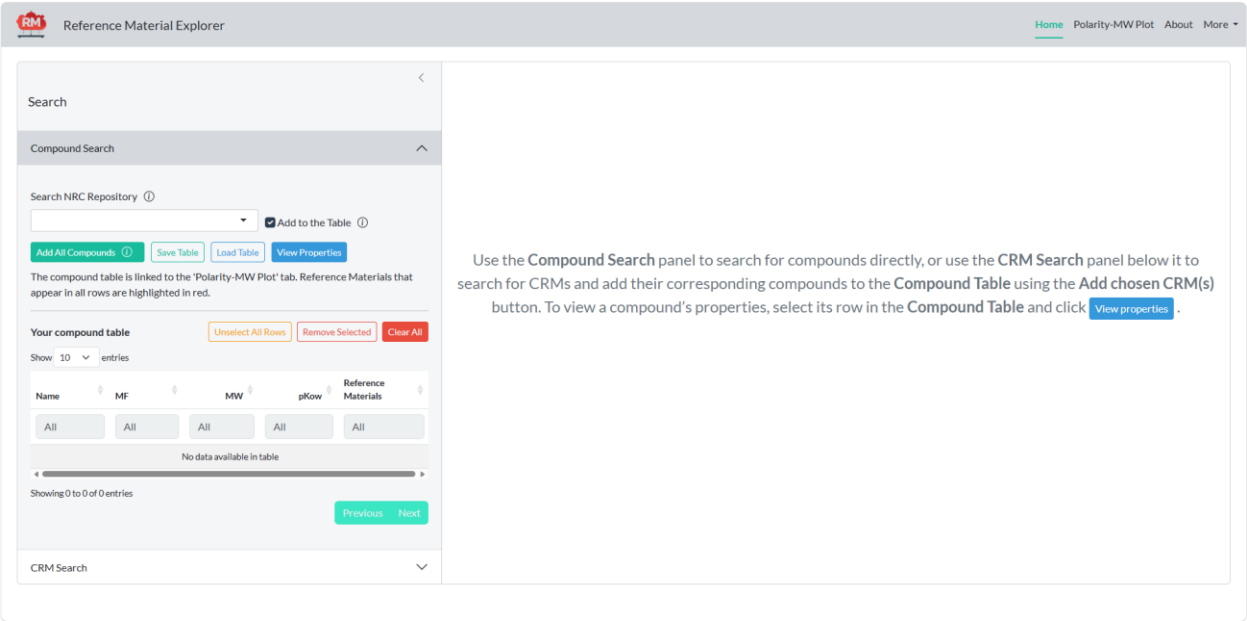
User Manual

Version 1.0

October 2026

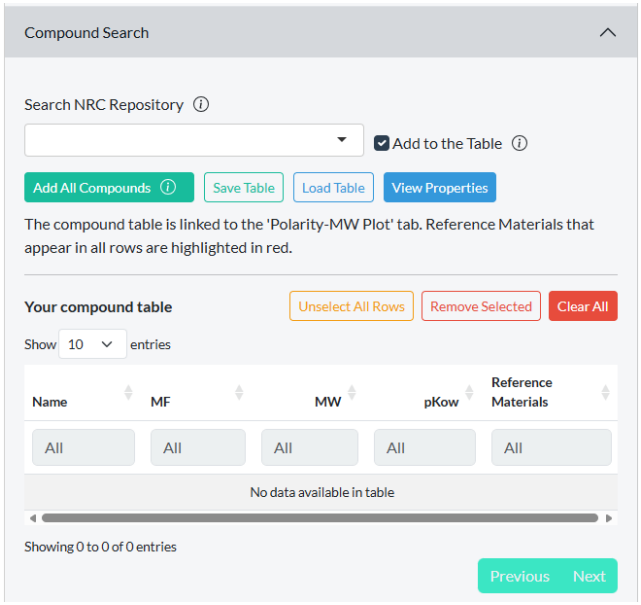
Home Page

The home page is where you can search for compounds and certified reference materials (CRMs). It contains a sidebar with two tabs: **Compound Search** and **CRM Search**. The main area of the home page displays the properties of a selected compound. Use the arrow button at the top right of the sidebar panel to collapse or expand the sidebar.

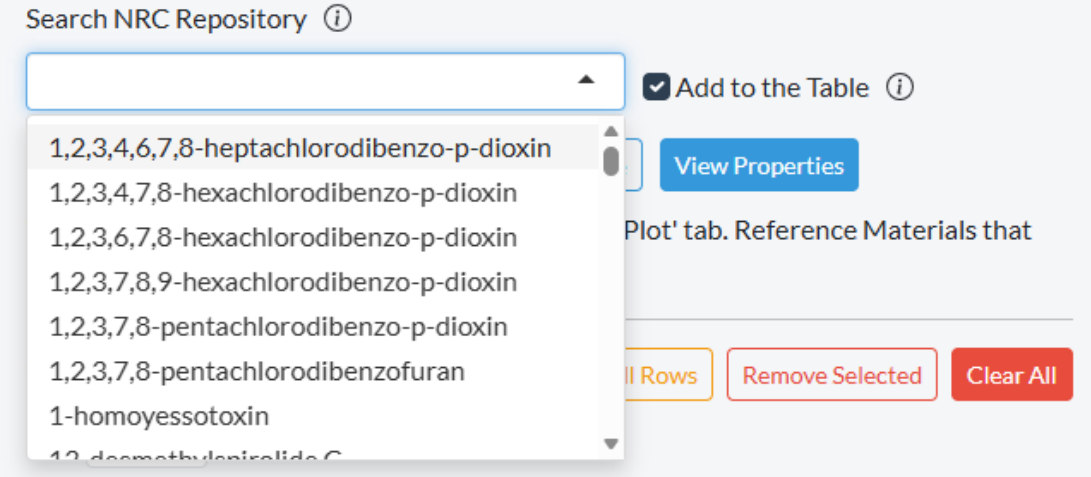


Compound Search

The **Compound Search** tab contains a search bar and a table to display compound information.

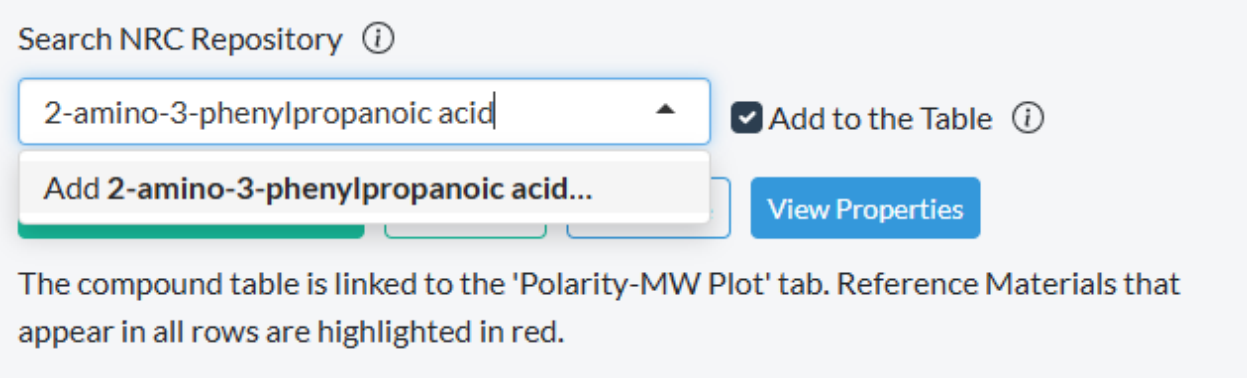


Search Bar: Type a compound name in the search bar. If the compound exists in the NRC repository, it will appear in the dropdown list for you to select and add to the table.

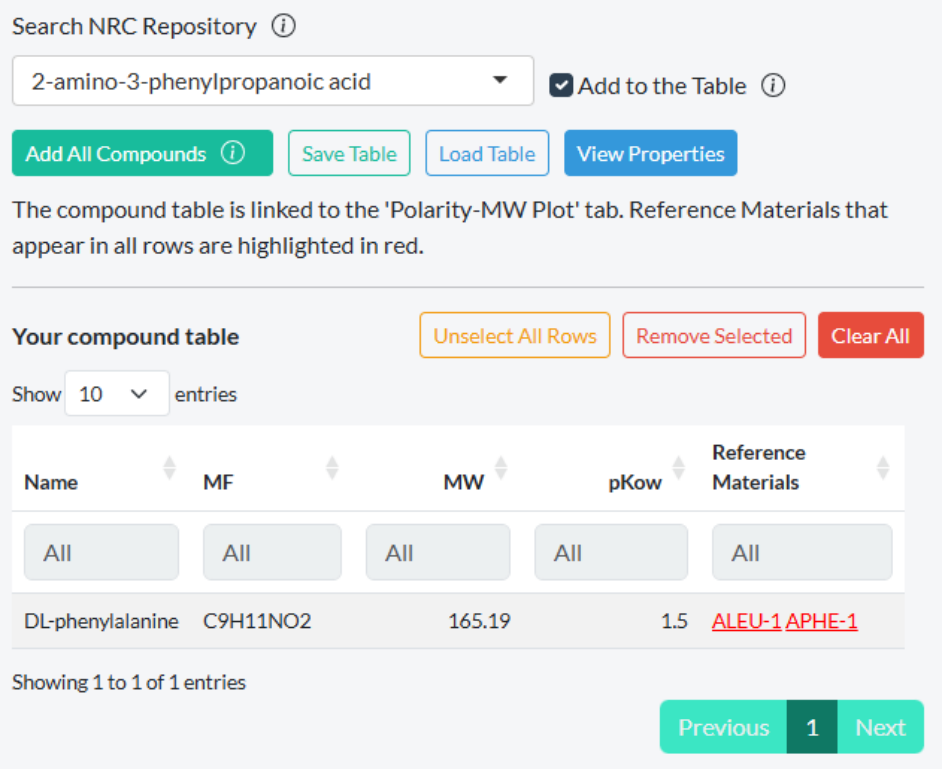


If the compound does not appear in the dropdown, an option to **Add [compound name]** will be displayed. Click this option to add the compound, and the following steps will occur automatically:

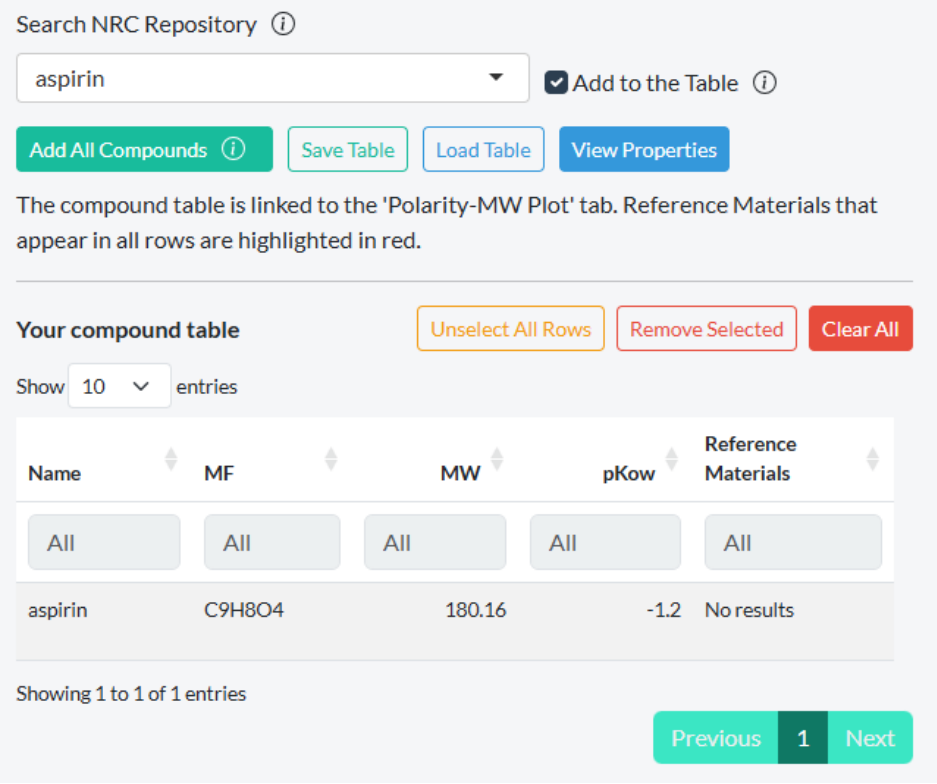
- **Synonym Resolved Search:** The compound will be searched by InChiKey to account for different synonyms
- **PubChem Search:** If the compound is still not found in the repository, it's properties will be retrieved from PubChem



Option to add compound



Synonym search: 2-amino-3-phenylpropanoic acid is listed in the NRC repository as DL-phenylalanine

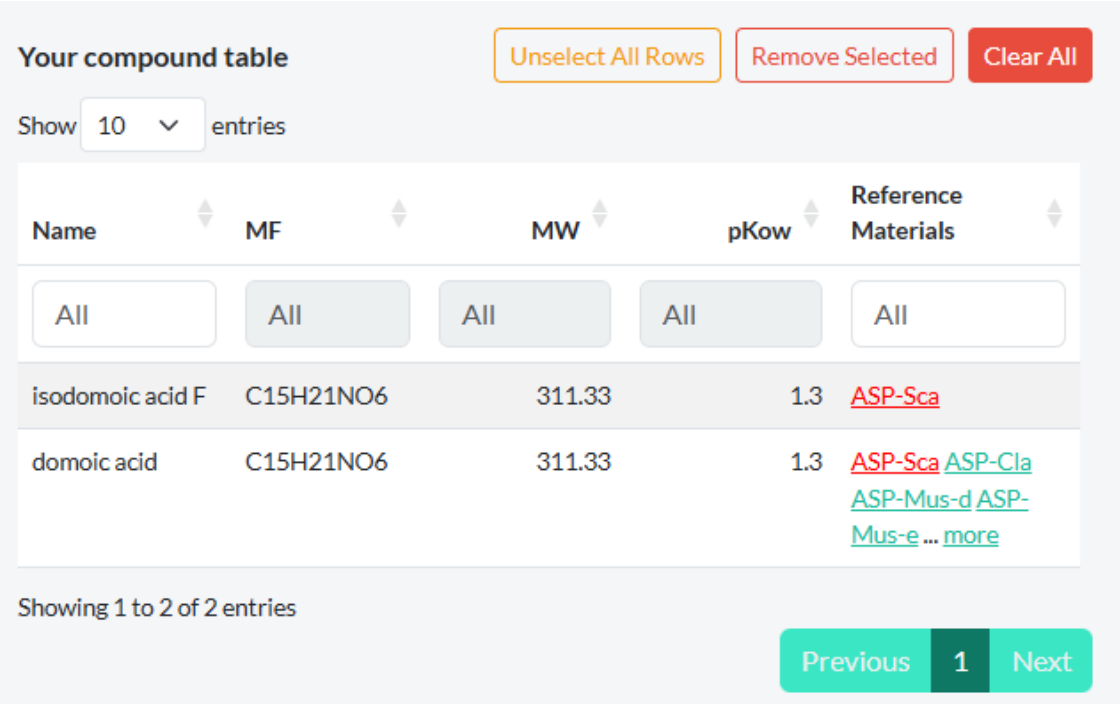


PubChem search: aspirin is not in the NRC repository; its properties were retrieved from PubChem and its reference materials are empty

Table Features:

- **Add to the Table:** When checked, searching for a new compound adds it to the existing table. Uncheck this box to clear the table before adding a new compound
- **Add All Compounds:** Adds all compounds in the NRC repository to the compound table
- **Save Table:** Saves the current compound table as a .csv file
- **Load Table:** Allows you to upload a previously saved .csv file
- **View Properties:** Select a compound in the table and click this button to view its detailed properties

Compound Table: The table displays information about each compound, including the CRMs in which it appears. CRMs that contain all the analytes in the table are highlighted in red. This can be used to search for CRMs containing multiple analytes.

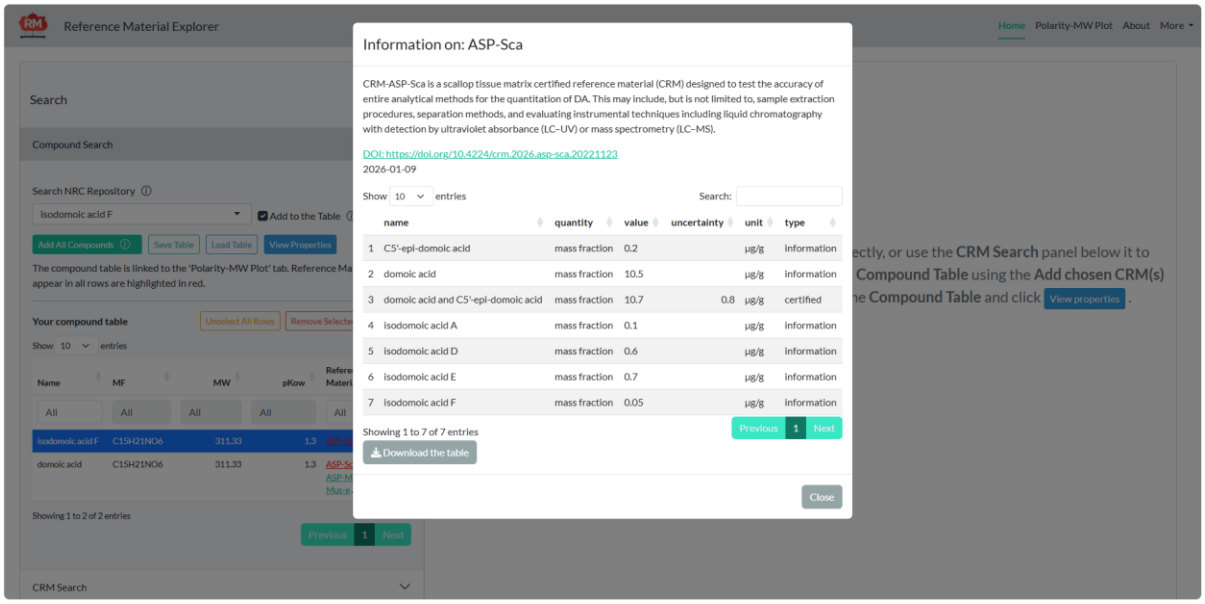


ASP-Sca is highlighted in red because it contains isodomoic acid F and domoic acid

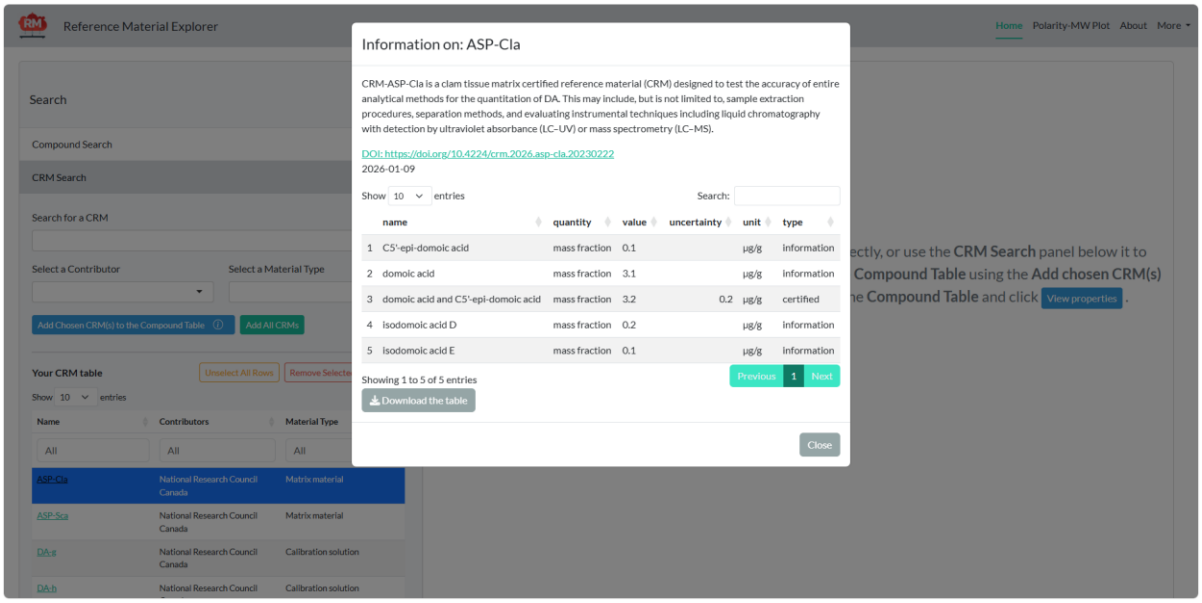
Click on a CRM to open a popup window containing:

- Summary information
- DOI
- Date
- Analyte table
- Button to download the analyte table as a .csv file

Note: CRM names are clickable throughout the app to access this popup information



Popup for ASP-Sca; accessed through the Reference Materials column in the compound table



Popup for ASP-Cla; accessed through the Name column in CRM table

Table Controls:

- **Unselect All Rows:** Click to unselect all highlighted rows
- **Remove Selected:** Select one or more compounds and click to remove them from the table
- **Clear All:** Removes all compounds from the table
- **Filters:** Use the search bars below each column header to filter the table

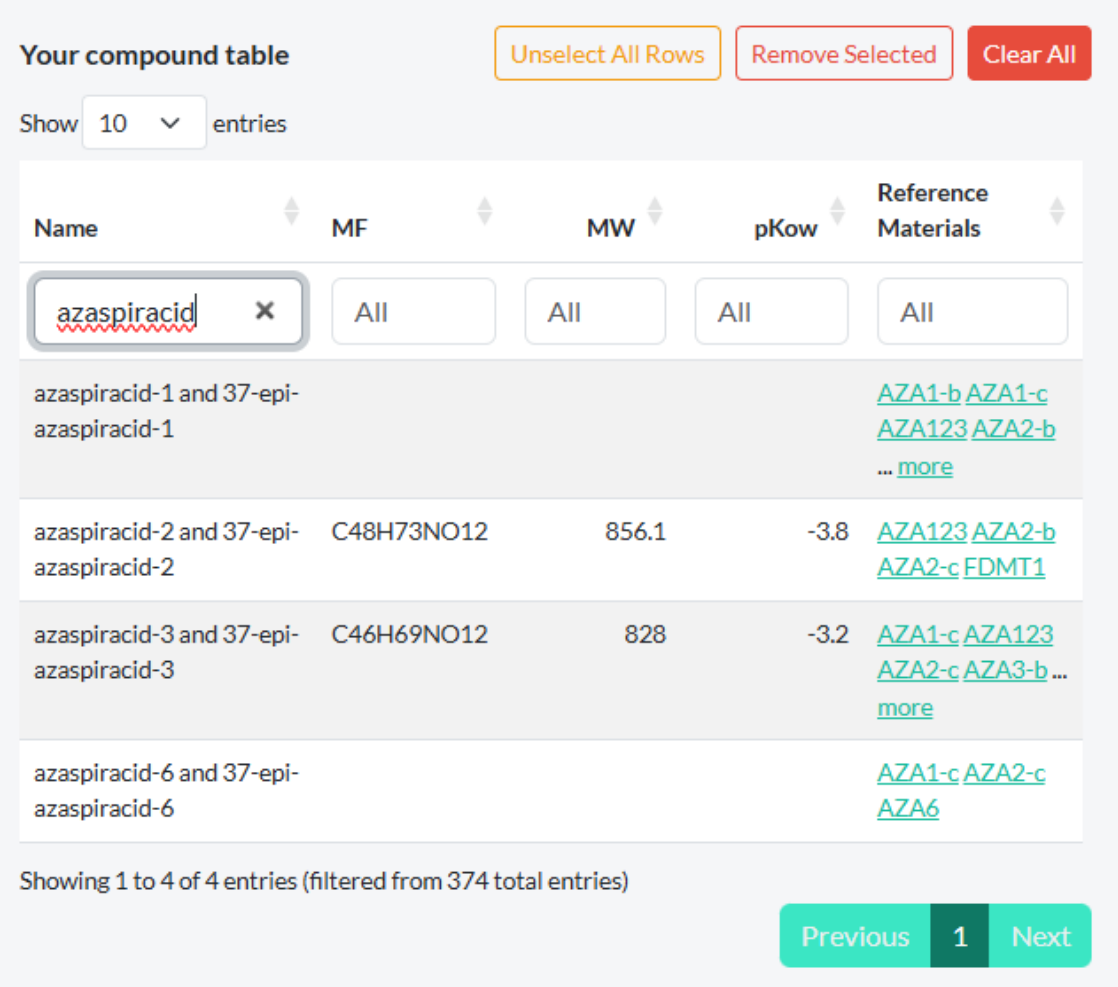


Table filtered by name

CRM Search

The **CRM Search** tab is located below the Compound Search tab in the side panel. It contains three search bars to find CRMs by:

- Name
- Contributor
- Material Type

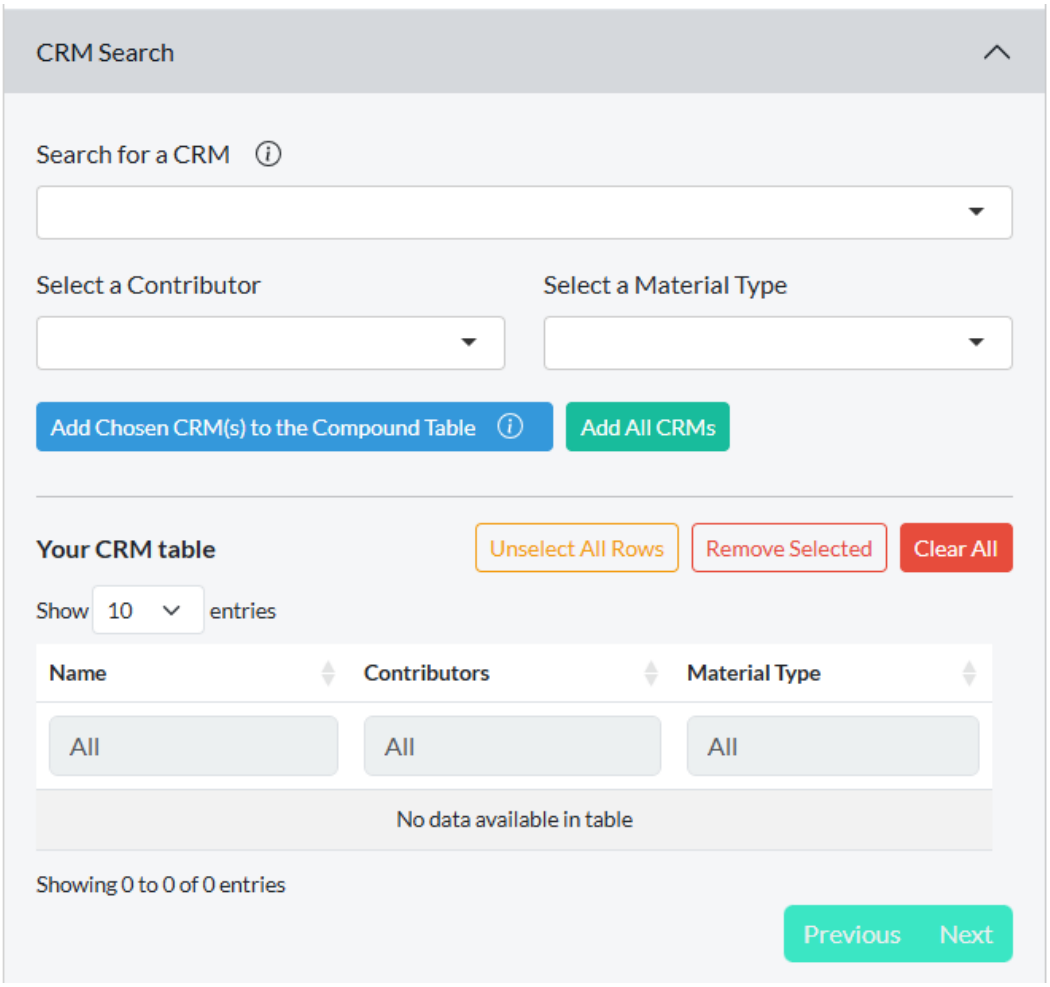
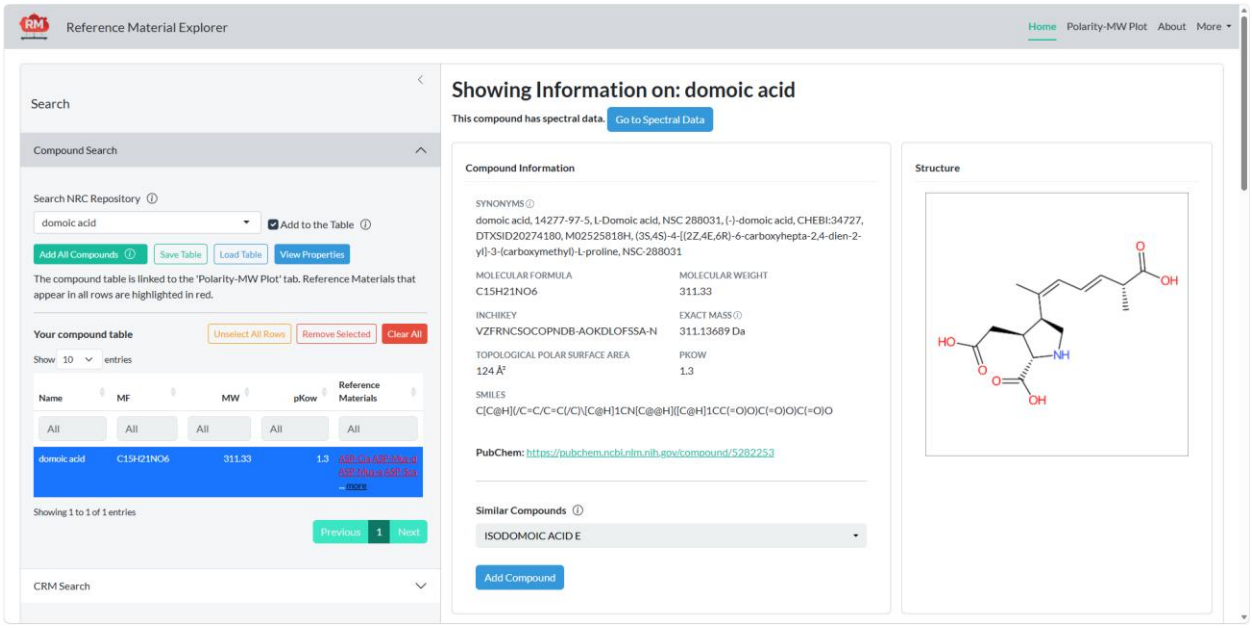


Table Features:

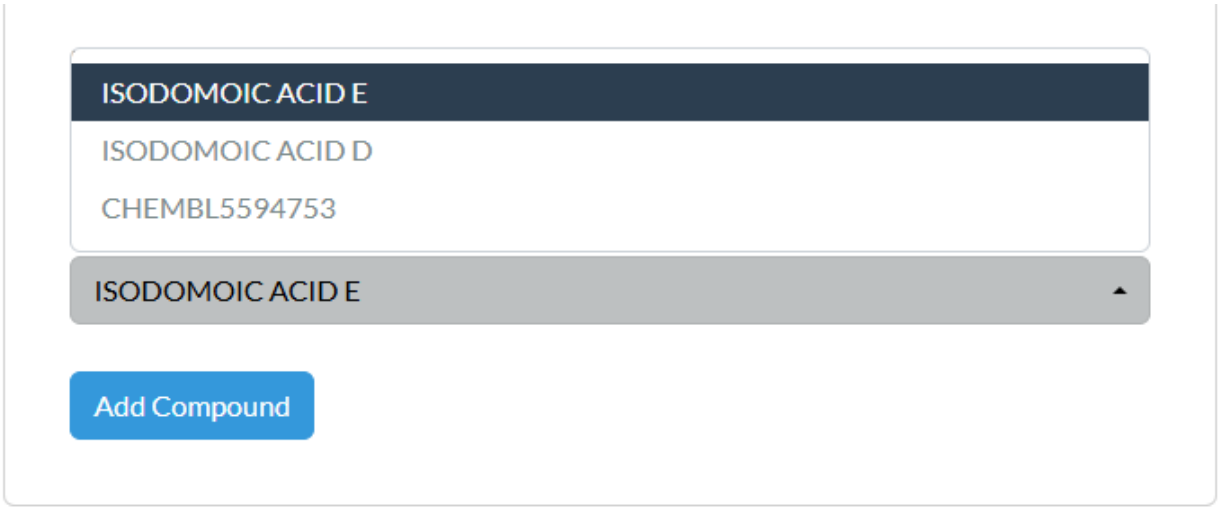
- **Add Chosen CRM(s) to the Compound Table:** Select one or more CRMs and click to add all compounds from those CRMs to the compound table
- **Add All CRMs:** Adds all CRMs from the NRC repository to the CRM table

Properties

The main area of the home page displays detailed properties of the selected compound. Click the **View Properties** button in the Compound Search tab to view a compound's properties.

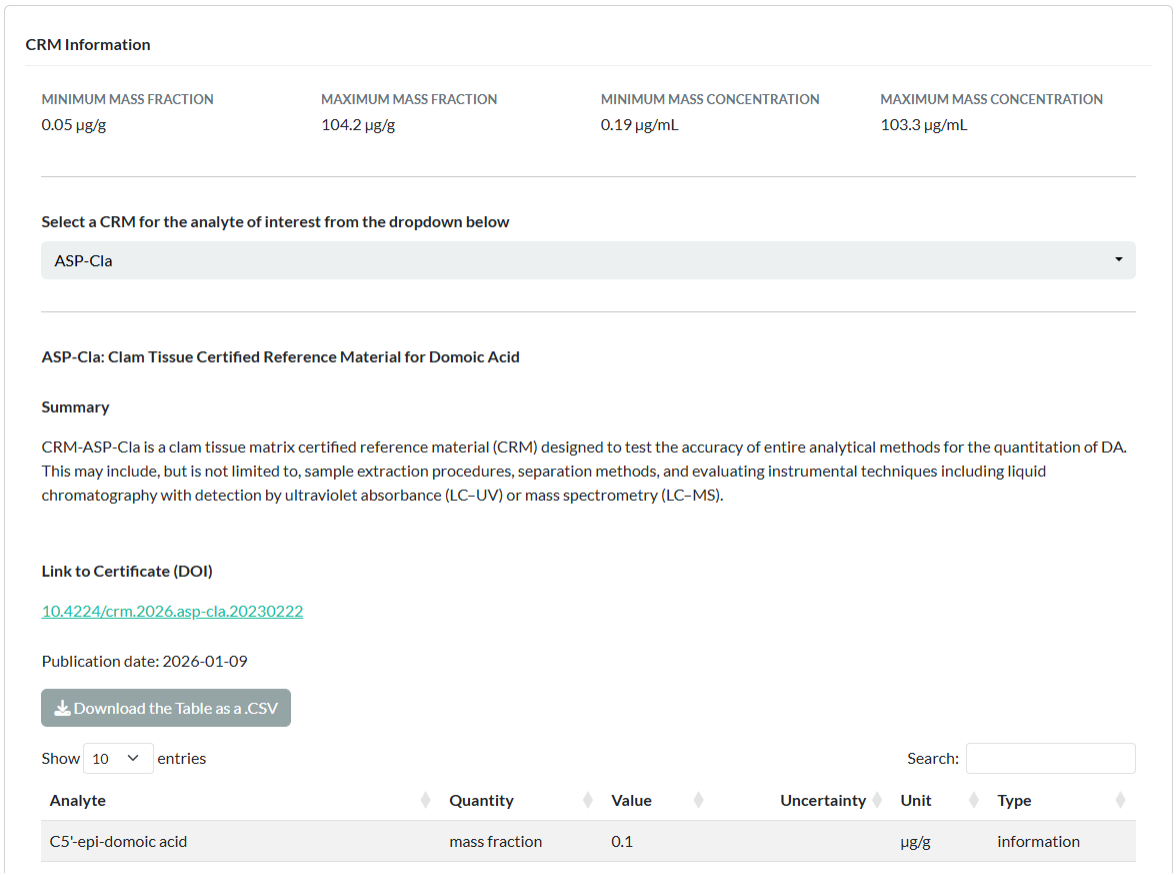


Similar Compounds: This will list similar compounds based on Tanimoto Coefficient if they are present in the NRC repository. Select a similar compound and click **Add Compound** to add it to the compound table.



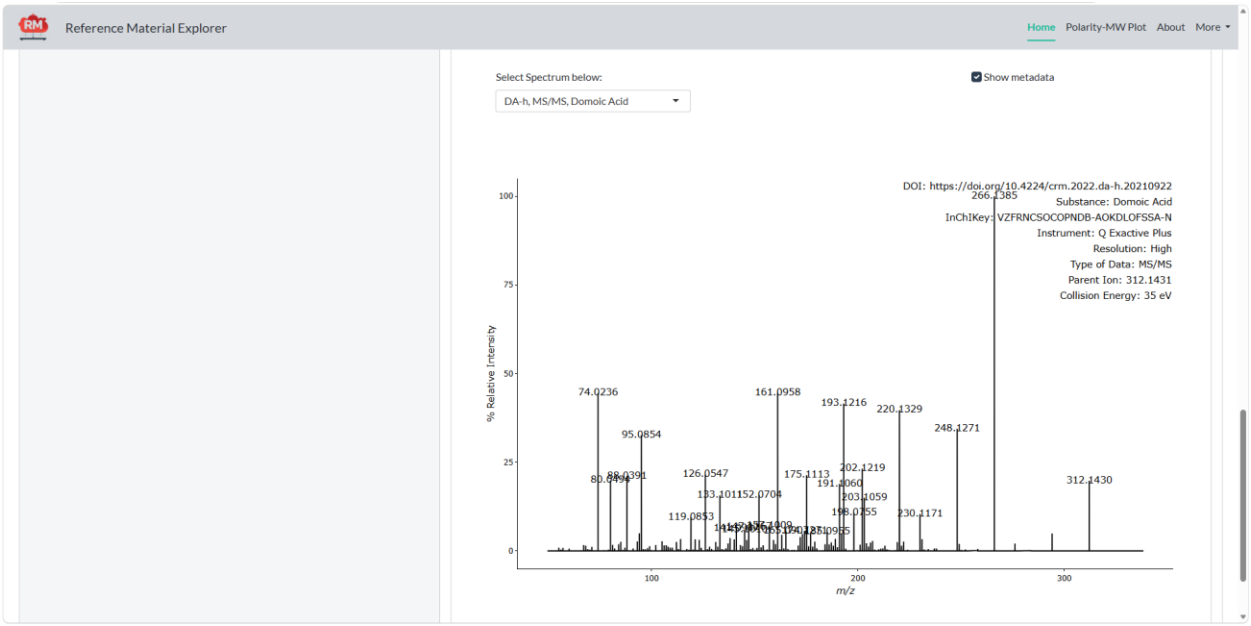
CRM Information: Scroll down the properties page to view the CRM Information card. This section displays:

- The minimum and maximum mass fraction and mass concentration values for the selected compound across all NRC CRMs
- A dropdown listing all CRMs the selected compound is featured in
- Information about the CRM selected in the dropdown including:
    - Summary information
    - DOI
    - Date
    - Analyte table
    - Button to download the analyte table as a .csv file

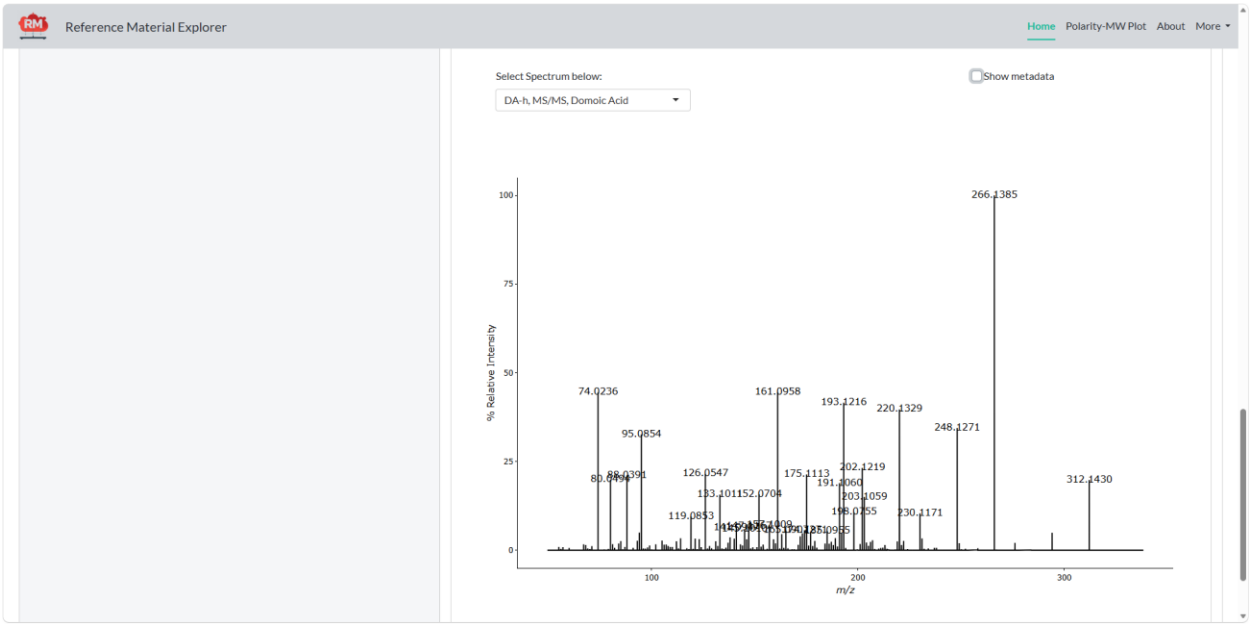


Spectral Data: Located at the bottom of the properties page. If spectral data is available for the selected compound:

- Click **Go to Spectral Data** (at the top of the page) to jump directly to this section
- Select the data type from the dropdown to display the plot
- **Show metadata:** Check this box to display metadata on the plot; uncheck to hide it
- **Download the Data:** Click this button to download the spectral data as a .csv file



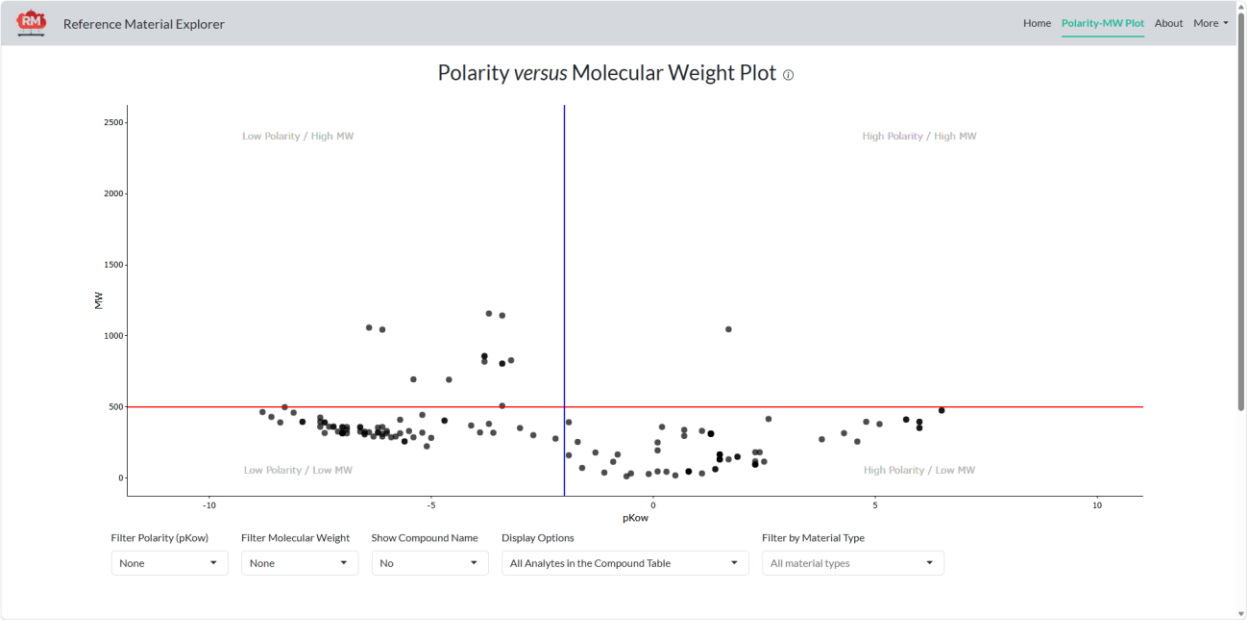
Spectral data plot with metadata shown



Spectral data plot with metadata hidden

Polarity-MW Plot

This plot displays compounds from the compound table that have both a pKow value (polarity) and a molecular weight value.

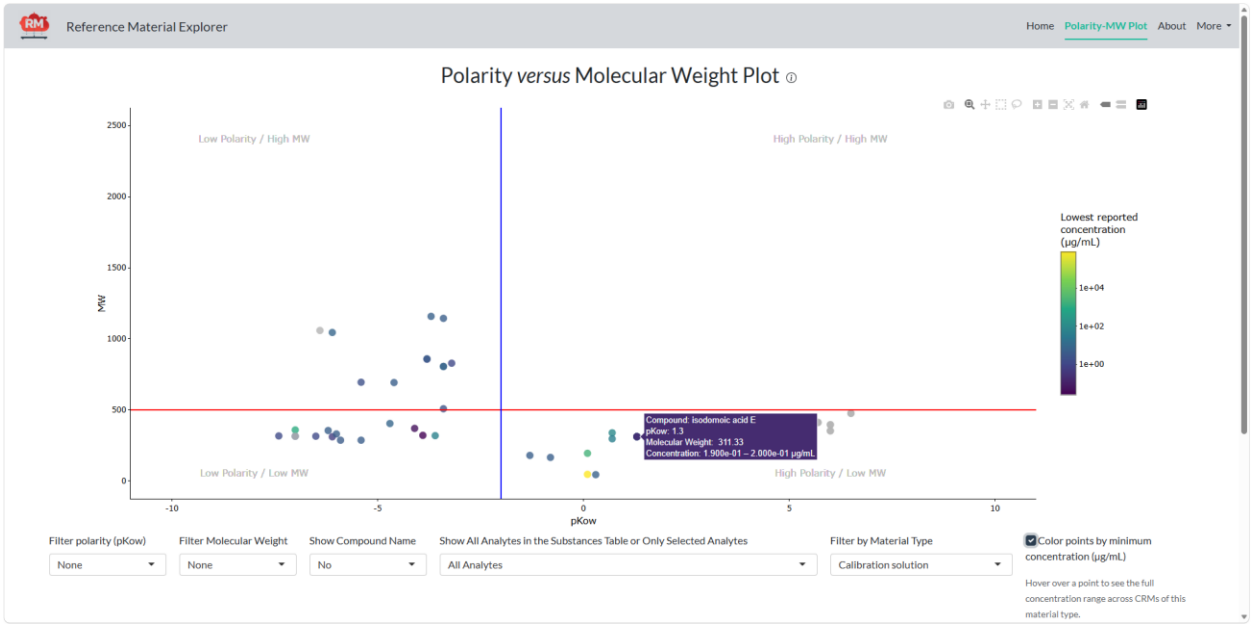


Filter Options:

- **Filter Polarity (pKow):** Select low polarity (≤ -2) or high polarity (> -2)
- **Filter Molecular Weight:** Select low molecular weight (≤ 500) or high molecular weight (> 500)
- **Show Compound Name:** choose “Yes” to display compound names on the plot
- **Display Options:**
    - **All Analytes in the Compound Table:** Displays all compounds from the compound table
    - **Only Selected Analytes:** Displays only the compounds you’ve selected in the table
- **Filter by Material Type:** filter compounds by
    - Matrix material
    - Calibration solution
    - High-purity material

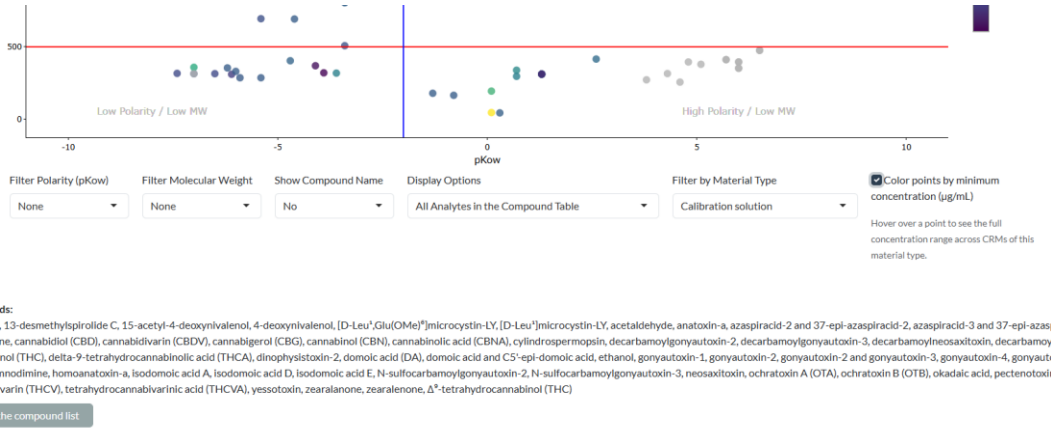
When a material type is selected, you can check **Color points by minimum concentration (µg/mL)** to:

- Display each compound colored according to its minimum concentration value across all NRC CRMs (points with no concentration data appear grey)
- Hover over a point to view the full concentration range



Data points colored by lowest reported concentration across all NRC reference materials

Filtered Compounds: Scroll down the page to see a list of compounds on the plot after filters have been applied. Click **Download the compound list** to download this list.



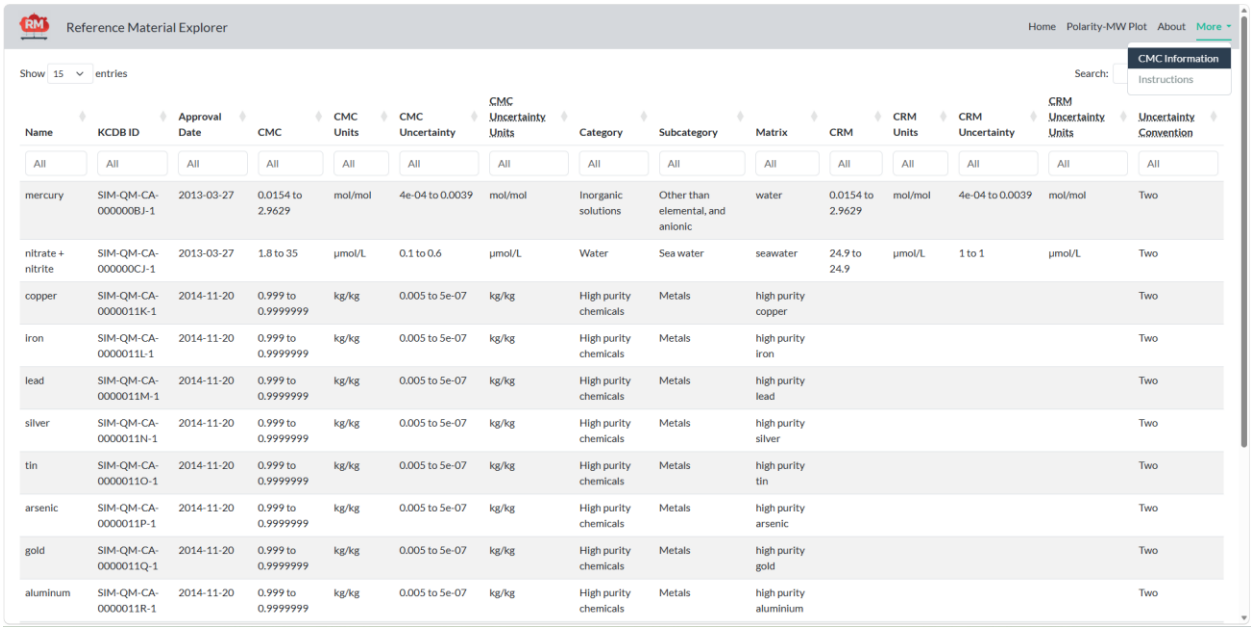
CMC Information

The CMC Information page can be found under the **More** tab in the navigation bar. It displays information from the KCDB API. Each column uses either **Uncertainty Convention One** or **Uncertainty Convention Two**, as specified by the last column.

Convention One is used when the expanded uncertainty range spans from the smallest numerical value of the uncertainty to the largest numerical value of the uncertainty found within the quantity range.

Convention Two is used when the expanded uncertainty range is expressed as the uncertainty of the smallest value of the quantity to the uncertainty of the largest value of the quantity; i.e., there is a link between the **'from'** entries and a link between the **'to'** entries for the dissemination range and the expanded uncertainty range.

You may hover over the **Uncertainty Convention** column header in order to view this description on the page.



Name	KCDB ID	Approval Date	CMC	CMC Units	CMC Uncertainty	CMC Uncertainty Units	Category	Subcategory	Matrix	CRM	CRM Units	CRM Uncertainty	CRM Uncertainty Units	Uncertainty Convention
All	All	All	All	All	All	All	All	All	All	All	All	All	All	All
mercury	SIM-QM-CA-000000BJ-1	2013-03-27	0.0154 to 2.9629	mol/mol	4e-04 to 0.0039	mol/mol	Inorganic solutions	Other than elemental, and anionic	water	0.0154 to 2.9629	mol/mol	4e-04 to 0.0039	mol/mol	Two
nitrate + nitrite	SIM-QM-CA-000000CJ-1	2013-03-27	1.8 to 35	µmol/L	0.1 to 0.6	µmol/L	Water	Sea water	seawater	24.9 to 24.9	µmol/L	1 to 1	µmol/L	Two
copper	SIM-QM-CA-0000011K-1	2014-11-20	0.999 to 0.9999999	kg/kg	0.005 to 5e-07	kg/kg	High purity chemicals	Metals	high purity copper					Two
iron	SIM-QM-CA-0000011L-1	2014-11-20	0.999 to 0.9999999	kg/kg	0.005 to 5e-07	kg/kg	High purity chemicals	Metals	high purity iron					Two
lead	SIM-QM-CA-0000011M-1	2014-11-20	0.999 to 0.9999999	kg/kg	0.005 to 5e-07	kg/kg	High purity chemicals	Metals	high purity lead					Two
silver	SIM-QM-CA-0000011N-1	2014-11-20	0.999 to 0.9999999	kg/kg	0.005 to 5e-07	kg/kg	High purity chemicals	Metals	high purity silver					Two
tin	SIM-QM-CA-0000011O-1	2014-11-20	0.999 to 0.9999999	kg/kg	0.005 to 5e-07	kg/kg	High purity chemicals	Metals	high purity tin					Two
arsenic	SIM-QM-CA-0000011P-1	2014-11-20	0.999 to 0.9999999	kg/kg	0.005 to 5e-07	kg/kg	High purity chemicals	Metals	high purity arsenic					Two
gold	SIM-QM-CA-0000011Q-1	2014-11-20	0.999 to 0.9999999	kg/kg	0.005 to 5e-07	kg/kg	High purity chemicals	Metals	high purity gold					Two
aluminum	SIM-QM-CA-0000011R-1	2014-11-20	0.999 to 0.9999999	kg/kg	0.005 to 5e-07	kg/kg	High purity chemicals	Metals	high purity aluminium					Two